

What is React?

- A JavaScript Library.
- used to Create Single Page application.
- Allows for the Creation of reusable UI Components.

Working of React



Why React?

- Manual DOM manipulation: traditional js use to change in directly modify the DOM, Slows down Performance.
- Complex State Management: Maintaining UI state became Messy and hard to debug.
- for e.g. →

import React from 'react'; → import React to Create Component.

function App() { → Define function

return (→ Return JSX.

<div>

<h1>hello, World! </h1>

</div>

);

}
export default App; → export the Component so
it doesn't use anywhere.



Working in React

(i)

React is JavaScript library, not a language.

- You write code in JS (with JSX SYNTAX).
- JSX lets you to write HTML-like code inside JS.

```
const element = <h1> Hello </h1>.
```

(ii)

JSX - Compiler to Real Javascript.

JSX is not valid JS. React uses Babel (a compiler) to convert JSX into pure JS.

```
<h1> Hello </h1>.
```

is to convert

```
act. createElement("h1", null, "Hello");
```

(iii)

React Builds the virtual DOM.

- React creates a virtual DOM.
- Instead of changing the real HTML DOM directly it changes this virtual version.

(vi)

React DOM. render() Mounts to Real DOM.

You Specify where in your HTML React should render:-

```
<body>
  <div id = "root"></div>
</body>
```

This Connects your JavaScript (React) code to your HTML Page via the id = "root" div.

```
import React from "react";
import ReactDOM from "react-dom";
```

```
ReactDOM.render(<App />, document.getElementById("root"));
```

✓

You write Components (Reusable UI Block)

```
function App() {
  return (
    <div className = "container">
      <h1> Hello </h1>
      <p> This is a component </p>
    </div>
  );
}
```

Here: (i) The App() function is a React Component

(ii) it results JSX (HTML inside JS).

(iii) This will be rendered inside the #root div in your HTML.

✓i

React updates the UI Automatically

React Uses State and Props to Manage data.

```
import React, {useState} from 'react';
```

```

function Counter () {
  const [count, SetCount] = useState(0);

  return (
    <div>
      <P> Count: {count} </P>
      <button onClick = { () => SetCount (count+1)}>
        Increase </button>
    </div>
  );
}

```

- When SetCount() is called, React automatically updates the virtual DOM.



Application of React

- Web Development
- Mobile app
- Enterprise Application
- Dashboards and Data Visualization.



New Features React 19

- Server-Side Rendering
- React Suspense Advancement
- Concurrent Mode
- Automatic Batching Enhancement
- New Hooks API